

Year 10 Measurement: Formula Reference

Page 1 of 2: Foundational 2D Area Formulas

Name: _____ Class: _____ Date: _____

Foundational 2D Areas

As we derive each formula in class, fill it into the right-hand column. The hint shows which variables to use.

Shape	Variables	Area Formula (fill in)
Square	side length: s	(uses s)
Rectangle	length: l , width: w	(uses l , w)
Triangle	base: b , height: h	(uses b , h)
Parallelogram	base: b , height: h	(uses b , h)
Trapezium	parallel sides: a and b , height: h	(uses a , b , h)
Rhombus	diagonals: d_1 and d_2	(uses d_1 , d_2)
Kite	diagonals: d_1 and d_2	(uses d_1 , d_2)
Circle	radius: r	(uses π , r)
Sector of a circle	radius: r , angle: θ (degrees)	(uses θ , π , r)
Annulus (ring)	outer radius: R , inner radius: r	(uses π , R , r)

Perimeter & Circumference

Shape	Variables	Formula (fill in)
Rectangle (perimeter)	length: l , width: w	(uses l , w)
Circle (circumference)	radius: r (or diameter: d)	(uses π , r or π , d)
Arc length	radius: r , angle: θ (degrees)	(uses θ , π , r)

Pythagoras' Theorem (right-angled triangles)

Rule	Variables	Formula (fill in)
Pythagoras	legs: a , b ; hypotenuse: c	(uses a , b , c)

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Page 2 of 2: 3D Surface Area & Volume

Surface area is the total area of all faces. Volume measures the space inside. Fill in each formula as we derive it.

Volume Formulas

3D Shape	Variables	Volume Formula (fill in)
Prism (general)	area of cross-section: A, length: l	(uses A, l)
Cube	side length: s	(uses s)
Rectangular prism	length: l, width: w, height: h	(uses l, w, h)
Cylinder	radius: r, height: h	(uses π , r, h)
Pyramid (general)	base area: A, perpendicular height: h	(uses A, h)
Cone	radius: r, perpendicular height: h	(uses π , r, h)
Sphere	radius: r	(uses π , r)

Surface Area Formulas

3D Shape	Variables	Surface Area Formula (fill in)
Cube	side length: s	(uses s)
Rectangular prism	length: l, width: w, height: h	(uses l, w, h)
Cylinder (closed)	radius: r, height: h	(uses π , r, h)
Cylinder (open / no lid)	radius: r, height: h	(uses π , r, h)
Cone (closed)	radius: r, slant height: l	(uses π , r, l)
Cone curved surface only	radius: r, slant height: l	(uses π , r, l)
Sphere	radius: r	(uses π , r)
Pyramid / prism (general)	sum of all face areas	(add all faces)